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Machine Learning based models for joint PDF shapes for multi-scalar mixing in turbulent flows

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Abstract: Supervised Machine Learning (ML) techniques are used to construct models for the joint Probability Density Function (PDF) of multi-scalar turbulent mixing. Data from non-reacting DNS calculations of three-scalar mixing in forced isotropic turbulence [1] are used to train models for the joint PDF. Data for the joint PDF, which evolves in time, from eleven different initial conditions, representative of a wide range of mixing states typically observed in practical combustion systems, are included in the training procedure. Traditional ensemble ML methods such as random forest and a fully connected feed forward Deep Neural Network are investigated. Results from *a-priori* analysis are compared against exact PDF shapes from the non-reacting DNS data with additional comparisons to various typically utilized distributions including the Dirichlet, the Connor-Mosimann (CM), Bi-variate Beta (BVB5), and statistically most likely distribution (SMLD). The advantages and disadvantages of the ML based models are discussed at the end.

Keywords: *Machine Learning (ML), Joint-PDF, LES*

1. Introduction

One of the primary modeling uncertainties associated with the presumed PDF class of models for reacting flows is the shape of the presumed PDF. Typically this PDF is parameterized by means and variances of the relevant scalars and is usually approximated using empirical hypotheses. For non-reacting passive scalars, such as mixture fraction, the beta distribution is the most popular choice for a marginal distribution and has been evaluated by a number of studies [2, 3]. For multi-scalar mixing the choice of the analytical model for representing the joint PDF is not still not clear. Doran *et al.*, Juneja and Pope, Sawford *et al.* and Cai *et al.* [4–7] are some of the previous studies that looked into multi-scalar mixing. Recently, Perry and Mueller [1] performed DNS of three component mixing in isotropic turbulence with eleven different scalar configurations and developed a systematic framework for selecting an appropriate form of joint PDF.

Instead of looking for an analytical model, this study focusses on using supervised learning based machine learning techniques to train models for the joint PDF of two passive scalars, $P(Z_1, Z_2)$. These ML based models will be trained using three stream mixing non-reacting DNS data from Perry and Mueller [1] and will be compared against the analytical models in reproducing

the joint PDF from DNS. Similar strategy was used by Frahan *et al.* [8] to construct a model for joint PDF for mixture fraction and progress variable.

2. Training Data

The data used to train and test all of the ML models for the joint PDF shapes are from the DNS of three-scalar mixing in forced isotropic turbulence [1]. In this DNS database, 11 different scalar configurations are considered. These configurations are representative of mixing states typically observed in practical combustion systems. The details regarding each of these scalar configurations can be found in [1] and will not be repeated here for brevity. Each case was computed on a mesh comprised of 256^3 cells. The background turbulent flow field was initialized at Taylor-scale Reynolds number, $Re_\lambda = 83$ with an eddy turnover time scale of 0.2 s. The DNS simulation was evolved in time with a time step of $50\mu s$. All the 11 scalar configurations are listed in Table 1.

Case	Configuration	$\tilde{Z}_1$	$\tilde{Z}_2$
I1	Isotropic	0.33	0.33
I2	Isotropic	0.33	0.33
I3	Isotropic	0.2	0.6
I4	Isotropic	0.33	0.33
I5	Isotropic	0.33	0.33
L1	Layered	0.33	0.33
L2	Layered	0.2	0.6
PI1	Isotropic	0.33	0.33
PI2	Isotropic	0.33	0.33
PI3	Isotropic	0.33	0.33
PL1	Layered	0.33	0.33

Table 1: DNS cases from [1] considered for training the joint PDF shapes

For model training purpose, the entire 3D field was saved at a frequency of 0.01 s. The joint PDF shapes were generated by splitting the entire 3D domain into cubic domains, representing a box filter, comprised of 32^3 , 64^3 , and 128^3 cells. Periodic boundary conditions were utilized to properly populate each of the boxes. Joint PDF shapes were generated from each of the 11 scalar configurations and 102 time snapshots. PDF shapes were binned on 64×64 bins in the (Z_1, Z_2) space. The joint PDF shapes are characterized by $\tilde{Z}_1$, $\tilde{Z}_2$, $\tilde{Z}_1'^2$, $\tilde{Z}_2'^2$, and $\widetilde{Z_1'Z_2'}$, which will be referred to as input variables. For each ML technique discussed in Sec. 3, two sets of models will be evaluated:

1. Means and variances as input variables: $P(Z_1, Z_2; \tilde{Z}_1, \tilde{Z}_2, \tilde{Z}_1'^2, \tilde{Z}_2'^2)$.
2. Means, variances, and co-variance as input variables: $P(Z_1, Z_2; \tilde{Z}_1, \tilde{Z}_2, \tilde{Z}_1'^2, \tilde{Z}_2'^2, \widetilde{Z_1'Z_2'})$.

Before training the model, all input parameters are scaled using the Robust scaler from the *scikit learn* library [9]. Robust scaler centers and scales each of the input variables independently and is found to work effectively on datasets with several outliers.

For the purposes of training and testing the ML based model, joint PDF shapes from all scalar configurations, all snapshots in time, and three different box widths were concatenated and randomly shuffled. A total of 198,000 and 1650 joint PDF shapes were thereafter randomly selected for training and testing the model, respectively. All of the results shown in Sec. 4 are generated from 256^3 cells. None of these joint PDF shapes were used for training of the ML based models.

3. Machine Learning (ML) techniques

In this study two kinds of ML techniques are investigated: Random Forests (RF), an ensemble based method, and a feed forward deep neural network (DNN). RF is a popular ML technique known for its flexibility and out-of-the-box accuracy. During training, RF constructs a forest of decision trees, where branches of each tree are the input features. The end points of these decision trees are the leaves which hold the output variables, otherwise known as the target values. During the model prediction phase, RF provides an average of the ensemble of these decision trees. The two key parameters that control the accuracy of RF techniques are the number of decision trees and the depth of each tree. In this work, the RF model is constructed using the *scikit learn* library [9], with 100 decision trees and a maximum tree depth of 30 nodes, resulting in a model size of 48 GB. Increasing the number of decision trees or the tree depth would increase the accuracy of the model but at the expense of increasing the cost of prediction loop.

The second ML technique considered in this work is deep neural network (DNN). In the last decade, improvements in algorithms and computing technology have enabled phenomenal success in the areas of computer vision, image processing, speech recognition, and natural language processing [10–12]. In this study a simpler feed-forward, fully connected DNN will be used to construct a model for the joint PDF. The network consists of three hidden layers and an output layer. The hidden layers have 256, 512 and 2048 fully connected nodes and a leaky rectified linear unit activation function. The entire training data are cycled over 1500 epochs. In each epoch cycle training data is shuffled and divided into batches of 64 samples. The Adam optimizer, an extension of the stochastic gradient descent algorithm, is used to update weights in the network. An initial learning rate of 10^{-4} was used for training. The Pytorch [13] library was used to build the network and was trained on a single NVIDIA Tesla K80 GPU.

4. Results and Discussion

4.1 Error Metrics

In order to evaluate the ML based model predictions, an error metric similar to the one introduced in [1] will be used in the current study. The L1 norm based error metric, ϵ is defined as

$$\epsilon = \sum_i \sum_j |P_{DNS}(Z_1^i, Z_2^j) - P_{ML}(Z_1^i, Z_2^j)| A_{ij} \quad (1)$$

where A_{ij} is the area of bin to used to discretized the joint PDF. Here i, j correspond to discrete bins in Z_1, Z_2 space.

4.2 Isotropic cases

Time evolution of the scalar mixing field from the DNS of the isotropic case, I1, is presented in Figs. 1 and 2. Joint PDF predictions from the ML based models are also shown. At early times, only three corner scalar states corresponding to pure components can be seen in Fig. 1. At later times, the joint PDF transitions into a triangle, corresponding to a partially mixed state, with a peak around the mean values. Eventually the joint PDF converges to a delta function at the prescribed mean values.

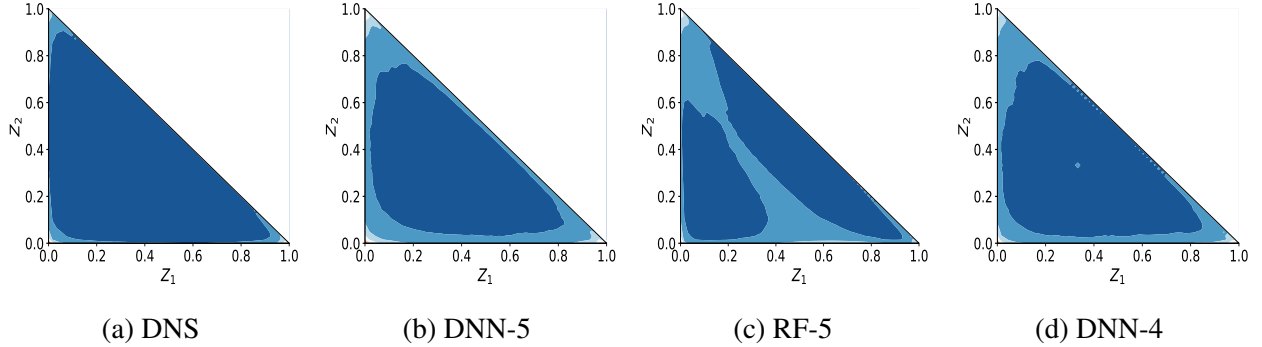


Figure 1: Case: I1 at $t/\tau_{eddy} = 0.4$

Deep Neural Network (DNN) models with four $(\tilde{Z}_1, \tilde{Z}_2, \tilde{Z}_1''^2, \tilde{Z}_2''^2)$ and five $(\tilde{Z}_1, \tilde{Z}_2, \tilde{Z}_1''^2, \tilde{Z}_2''^2, \tilde{Z}_1''\tilde{Z}_2'')$ input parameters, DNN-5 and DNN-4, respectively, qualitatively reproduce the scalar mixing field represented by the joint PDF shown in Figs. 1 and 2. At $t/\tau_{eddy} = 0.4$, DNN models predict a more mixed state when compared against the DNS. Joint PDF shape prediction from Random Forest model with five inputs, RF-5, shows significantly more mixing between pure components. Figure 2 shows the output from the same set of models and DNS at $t/\tau_{eddy} = 0.7$. Qualitatively, DNN-5 reproduces all of the features observed in the DNS whereas DNN-4 under predicts the peak. RF-5 predicts the peak value corresponding to the prescribed means correctly but also predicts non-negligible probabilities of pure components.

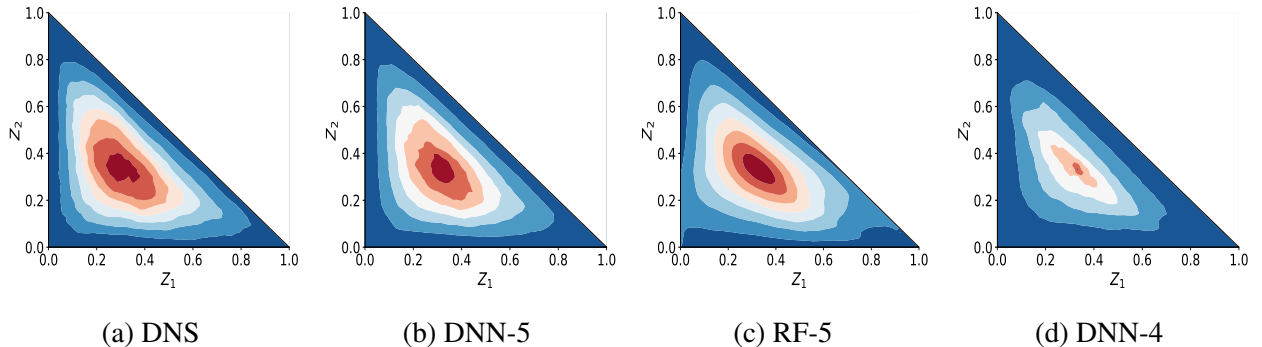
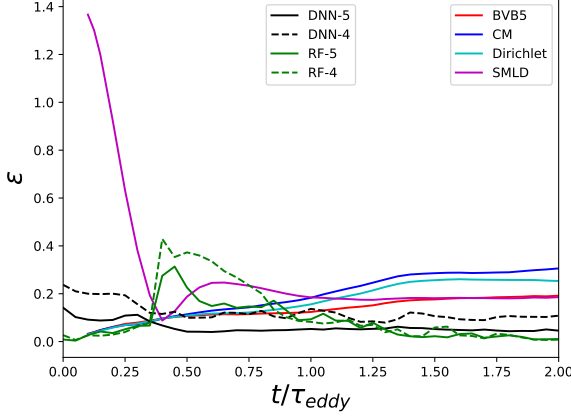


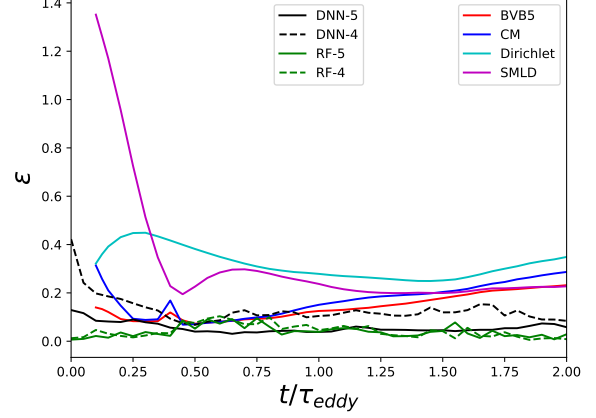
Figure 2: Case: I1 at $t/\tau_{eddy} = 0.7$

In order to make a more quantitative comparison between models, the time evolution of the error metric, ϵ , defined by Eq. 1, is presented in Fig. 3. Along with the predictions from ML based models, Fig. 3 also shows all of the analytical models described in [1]. Bivariate Beta,

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(a) I1



(b) I3

Figure 3: Time evolution of Error metric from Cases I1, I3

BVB5, and the statistically-most-likely distribution, SMLD, characterize the joint PDF shapes using five parameters, while Dirichlet and Connor-Mosimann, CM, use three and four parameters, respectively. More details on the formulation of these analytical forms can be found in [1] and references therein. For the same number of input parameters, DNN-5, predicts a lower error when compared to BVB5 and SMLD. Similarly, over two eddy turn over times, DNN-4 results are more accurate than Dirichlet and CM. Consistent with the trend observed in the contour plots in Fig. 1, both RF-4 and RF-5 predict a larger error at earlier times ($t/\tau_{eddy} \approx 0.5$) compared to analytical and DNN based models. At later times as the joint PDF shape approaches a delta function, RF based models reproduce joint PDF more accurately.

In contrast to the scalar configurations I1 and I2, Case I3 was constructed with unequal prescribed means, $\tilde{Z}_1 = 0.2$ and $\tilde{Z}_2 = 0.6$. Therefore, the joint PDF shown in Fig. 4 has a more skewed shape with a bias towards the $Z_2 = 1.0$ corner of the triangle. Similar qualitative trends in terms of predictions from ML based models is observed. Quantitative comparison of various analytical and ML models is shown in Figure 3b. RF based models, RF-5 and RF-4, predict lower error and are now comparable to the predictions made by DNN-4 and DNN-5. In terms of accuracy, ML based models tend to surpass the analytical models with same number of input parameters.

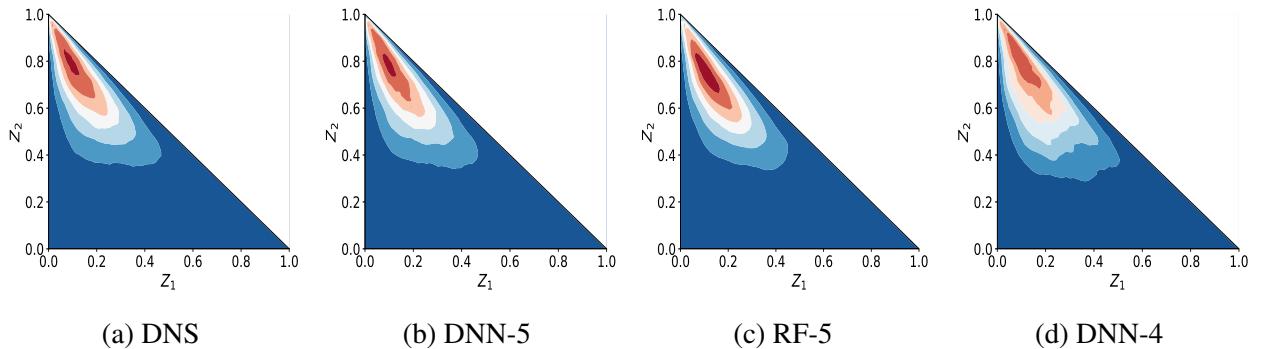


Figure 4: Case: I3 at $t/\tau_{eddy} = 0.7$

Joint PDFs for cases I4 and I5 are shown in Figs. 5 and 6. These cases force different mixing characteristics between components C_1 and C_3 by changing the initial length scales of scalars. Figures 5 clearly shows this effect while for case I4 the joint PDF is spread out in the Z_1 direction. In contrast to I4, the joint PDF from case I5 shows a more concentrated shape in the Z_1 direction. Qualitatively, all of the ML based models tend to over-predict the peak value at $Z_1 \approx 0.33$ and $Z_2 \approx 0.33$. RF based models erroneously predict a joint PDF shape qualitatively similar to the case I1. However, all ML based models accurately reproduce the joint PDF shapes from Case I5, shown in Figure 6. DNN-5 and RF-5 qualitatively compare accurately to the DNS results.

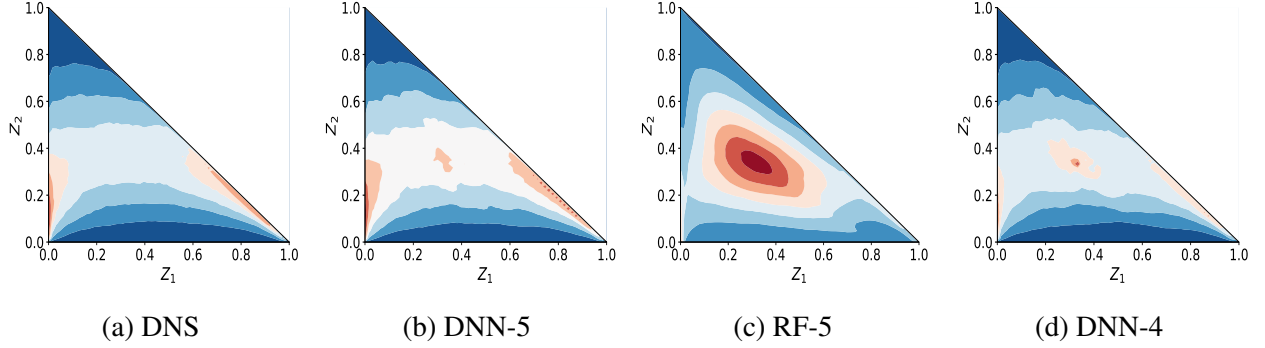


Figure 5: Case: I4 at $t/\tau_{eddy} = 0.4$

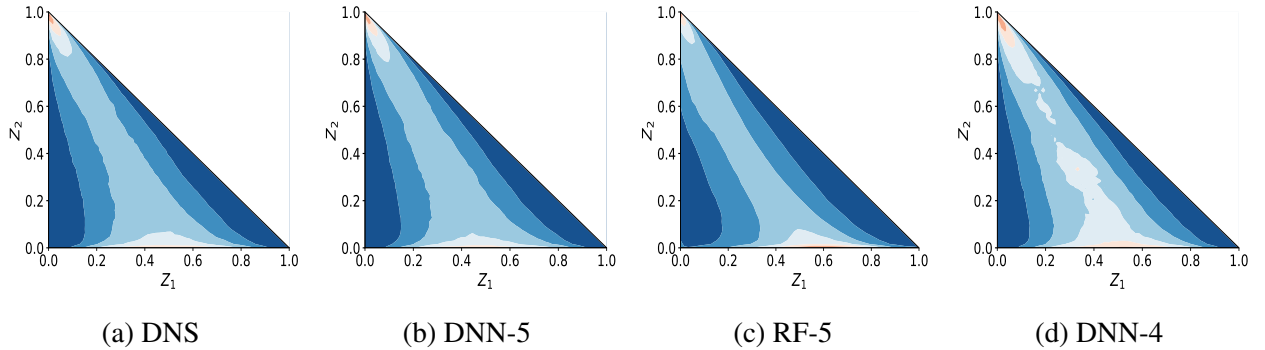
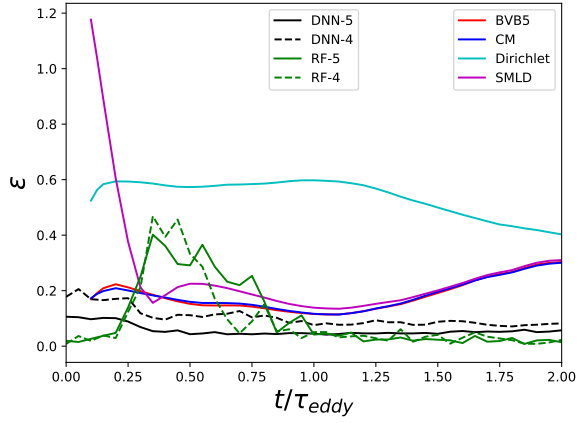


Figure 6: Case: I5 at $t/\tau_{eddy} = 0.4$

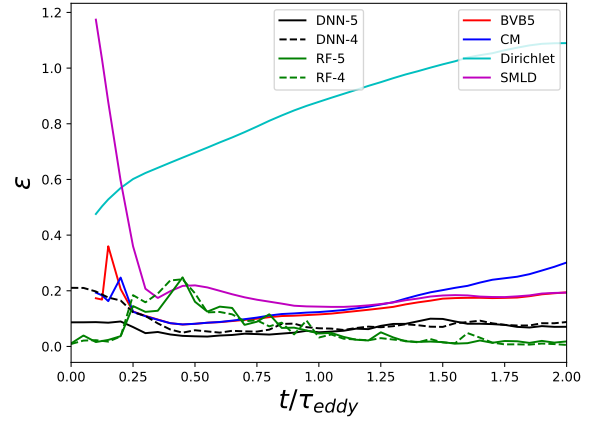
Time evolution of error metric, ϵ , for cases I4 and I5 are shown in Fig. 7. Over prediction of the peak value at $Z_1 \approx 0.33$, by RF based models, can be clearly seen from the error metric where the error sharply rises at $t/\tau_{eddy} \approx 0.4$. Perry *et al.* [1] had observed that BVB5 and CM-1 were able to fully capture the DNS trends. Fig. 7 show that both DNN-4 and DNN-5 produce errors smaller than these analytical models.

The Joint PDF shapes from cases PI1, PI2, and PI3 are presented are in Figs. 8, 9 and 10. As described in detail in [1], in these cases the components C_1 and C_3 are initialized in the same fashion as case I1 and are allowed to mix with increasing levels of premixing with the C_2 component initial condition from case I1 superimposed. The impact of increasing the level of premixing can be seen clearly seen in Figs. 8a, 9a and 10a, where the width of the joint PDF becomes narrower as premixing is increased. The ML based models qualitatively reproduce the joint PDF shapes from

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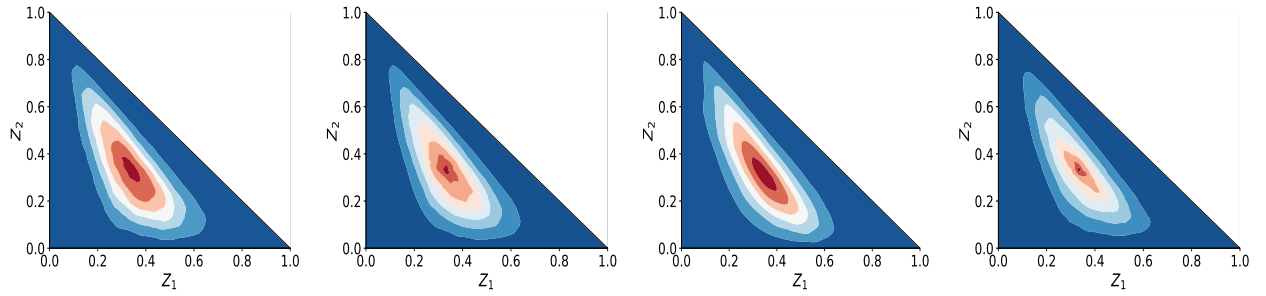
(a) I4



(b) I5

Figure 7: Time evolution of Error metric from Cases I4 and I5

DNS, as shown in Figs. 8, 9 and 10. More quantitative comparisons are shown in Fig. 11, where a consistent trend of superior performance by the ML based methods is noted.



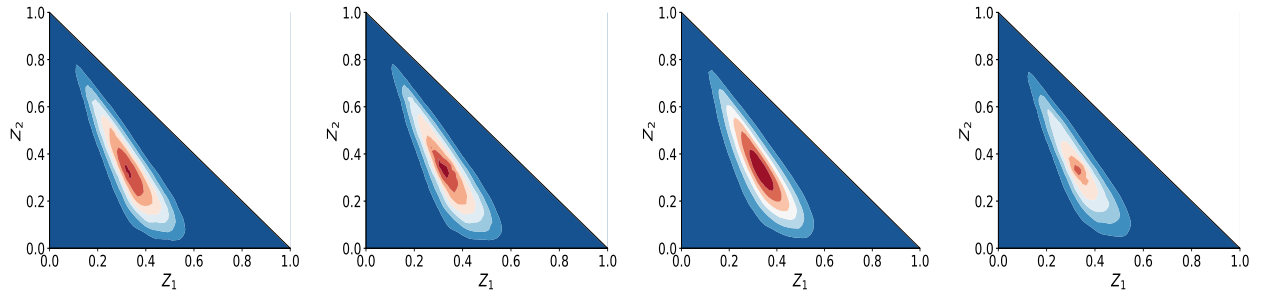
(a) DNS

(b) DNN-5

(c) RF-5

(d) DNN-4

Figure 8: Case: PI1 at $t/\tau_{eddy} = 0.7$



(a) DNS

(b) DNN-5

(c) RF-5

(d) DNN-4

Figure 9: Case: PI2 at $t/\tau_{eddy} = 0.7$

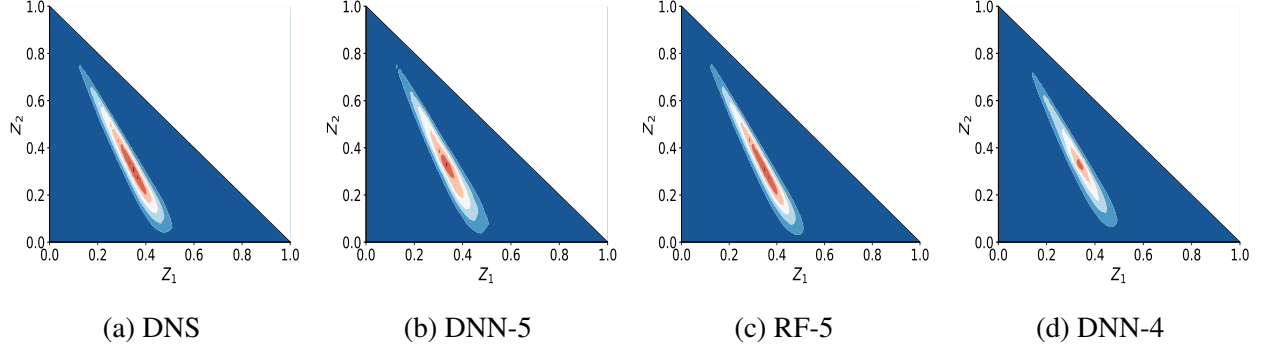


Figure 10: Case: PI3 at $t/\tau_{eddy} = 0.7$

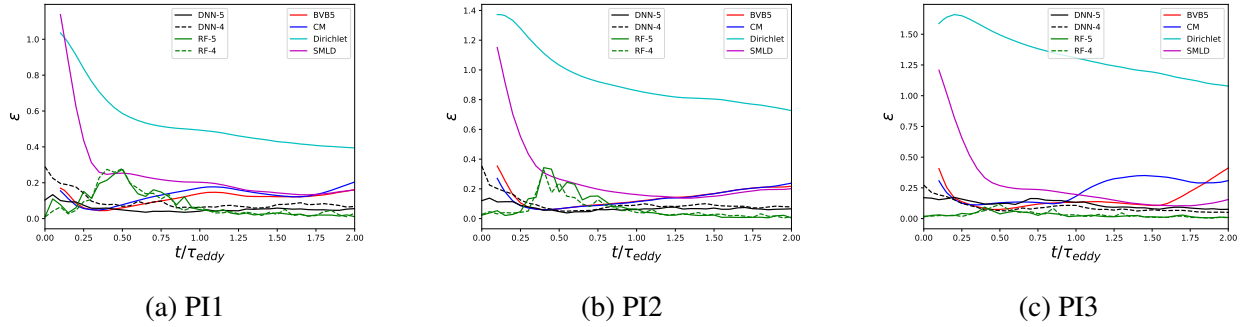


Figure 11: Time evolution of Error metric from Cases PI1, PI2 and PI3

4.3 Layered cases

Two layered cases, L_1 and L_2 , were also simulated in the DNS study [1]. These layered cases were designed to reproduce the scalar mixing configuration typically seen in co-annular jets. These simulation cases were distinguished by unequal (L_2) and equal (L_1) prescribed mean fields. Additionally, a partially premixed case was also designed by premixing the case L_1 for $0.5\tau_{eddy}$. As pointed out by [1] these layered test cases were the most challenging of all the configurations for the analytical models for joint PDF. The joint PDF shape from DNS and model predictions for the three cases (L_1 , L_2 and PL_1), at $t/\tau_{eddy} = 1.5$ are presented in Figs. 12, 13, and 14. The time variation of the error metric, ε , from ML and the analytical models is shown in Fig. 15. Amongst the analytical models, Perry *et al.* [1] had observed that BVBS throughout mixing process and SMLD at later times reproduced joint PDF from DNS most accurately. All of the other models were found to have relatively high error throughout the mixing process. Fig. 15 shows that DNN-4 and DNN-5 produced lower errors compared to the analytical models with the same number of input parameters, Dirichlet and CM, BVBS and SMLD, respectively. RF based models, regardless of number of input parameters, were observed to be accurate at earlier mixing times but produced relatively large errors around $t/\tau_{eddy} = 1.0$.

5. Conclusions

Machine Learning (ML) based techniques were used to construct models for the joint PDF of multi-scalar mixing, $P(Z_1, Z_2)$. ML techniques were trained using data from non-reacting DNS of three

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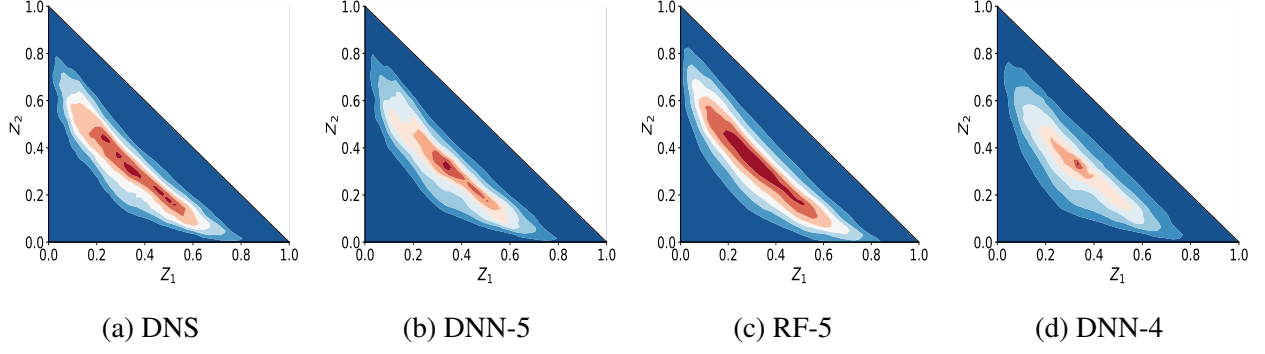


Figure 12: Case: L1 at $t/\tau_{eddy} = 1.5$

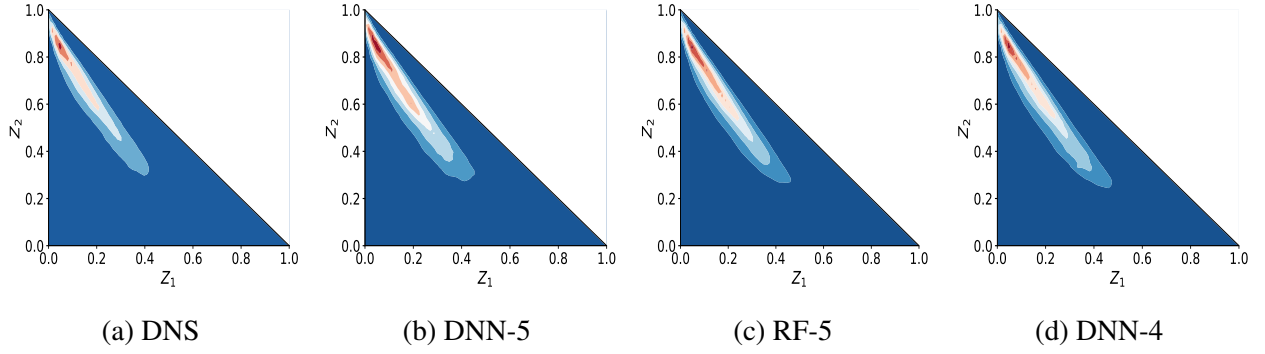


Figure 13: Case: L2 at $t/\tau_{eddy} = 1.5$

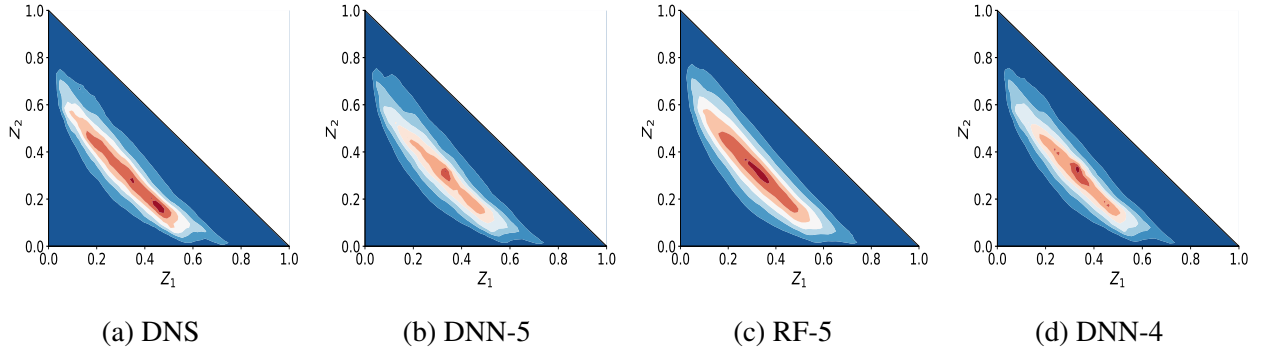


Figure 14: Case: PL1 at $t/\tau_{eddy} = 1.5$

scalar mixing. Time varying joint PDFs from eleven different initial conditions, representative of a wide range of mixing states, were used in the training procedure. Results from *a-priori* analysis demonstrated the accuracy of ML based models, in particular the deep neural network (DNN) model, in reproducing the joint PDF from all of the eleven scalar configurations. For the same number of input parameters, DNN based model surpassed the accuracy of all the analytical models. Future work would involve providing a subset of the DNS data for training the model to evaluate the predictive capability of these techniques in the areas where training data are sparse.

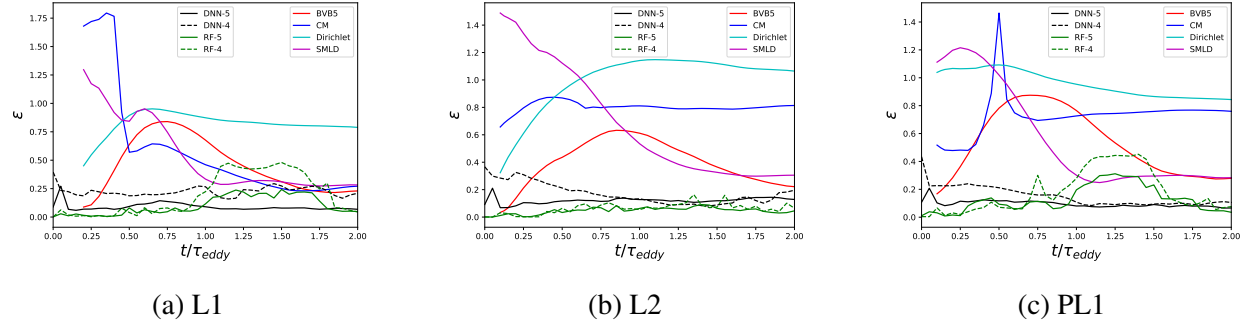


Figure 15: Time evolution of Error metric from Cases L1, L2 and PL1

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